

The key to understanding supersonic radiative Marshak waves experiments using simple models and advanced simulations



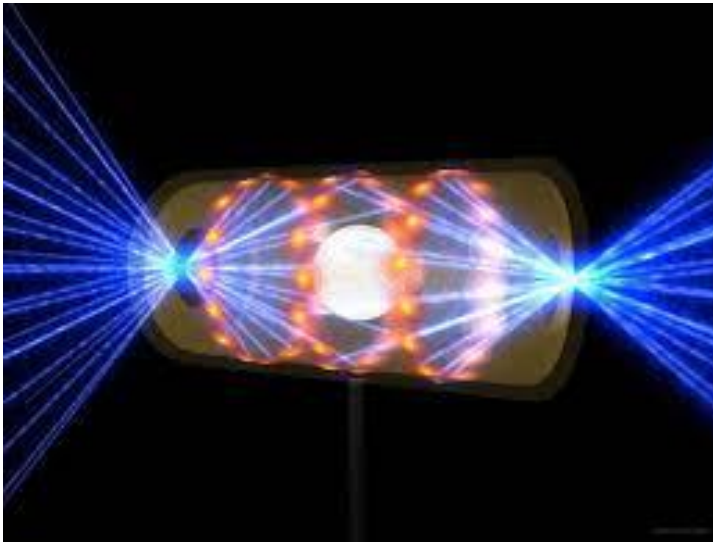
Avner P. Cohen, Shay I. Heizler

Physics department, *NRCN*, Israel.

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- Analytic Model
- Advanced simulations

Radiative heat waves

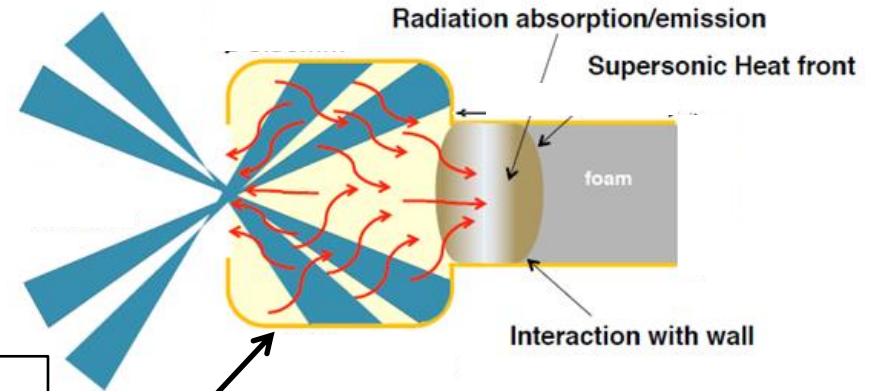
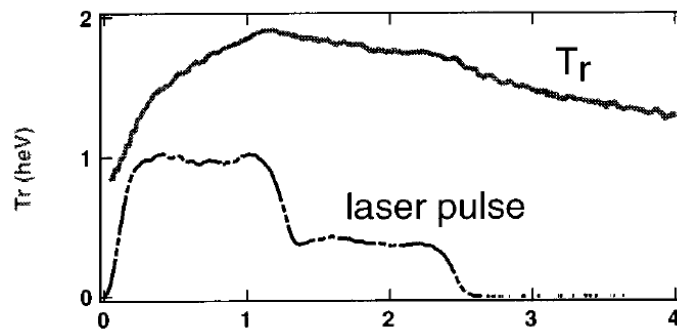


Inertial Confinement
Fusion (ICF) NIF



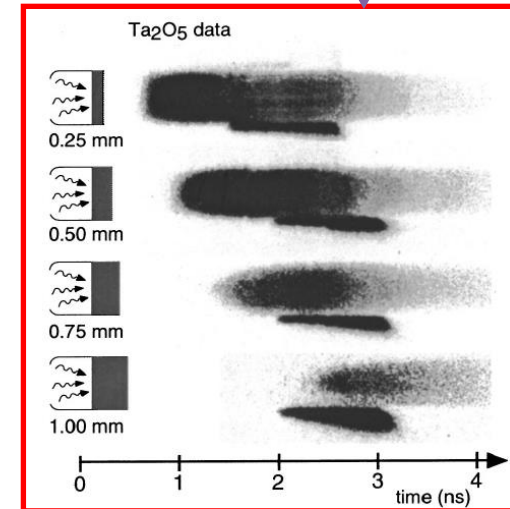
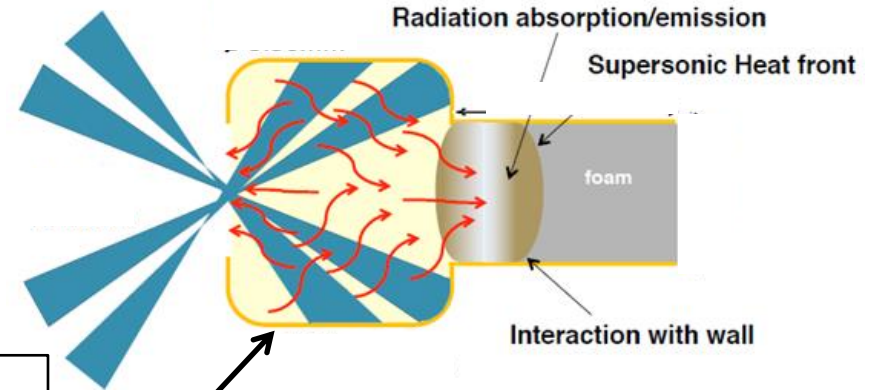
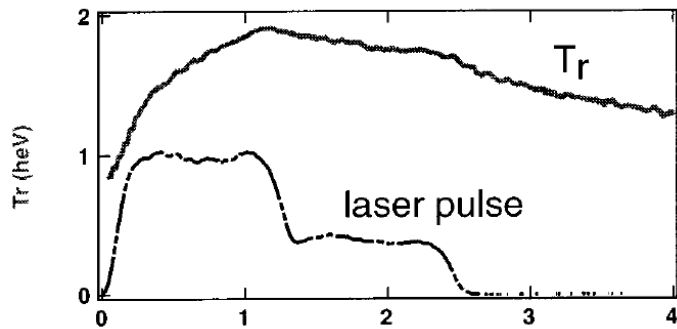
Astrophysics, energy
transport in stars

Marshak Waves Experiments Description

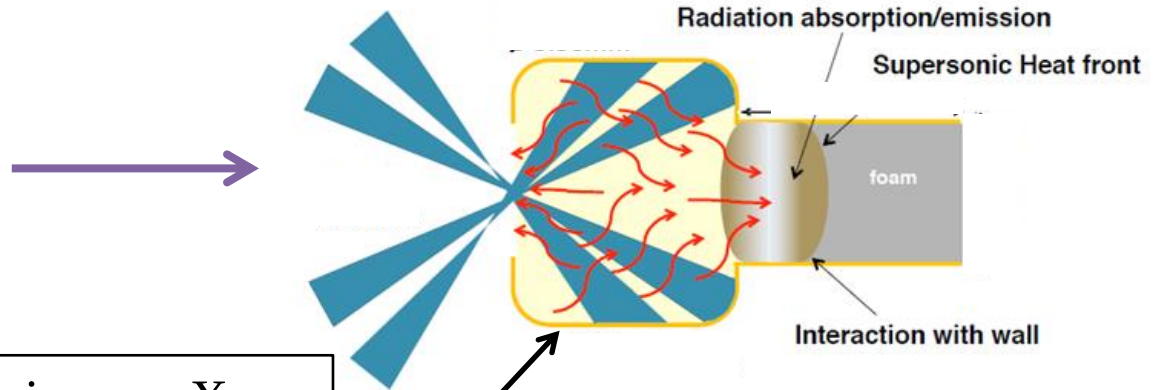
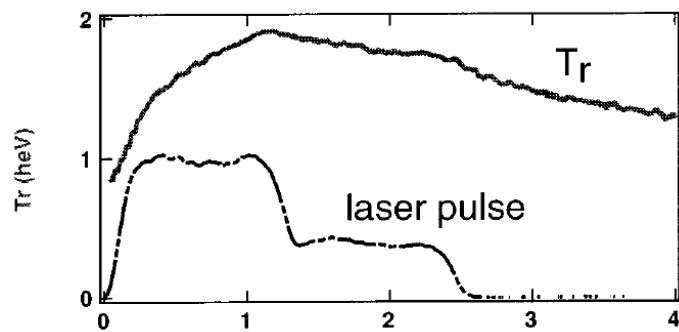


Using a “hohlraum”, as a primary X-ray generator, ($100 - 350\text{eV} \approx 1\text{-}4 \cdot 10^6 \text{ }^\circ\text{K}$), a supersonic heat-wave is propagated in the low-density foam

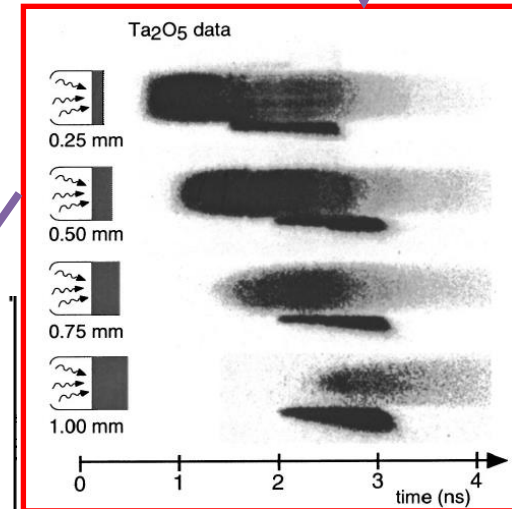
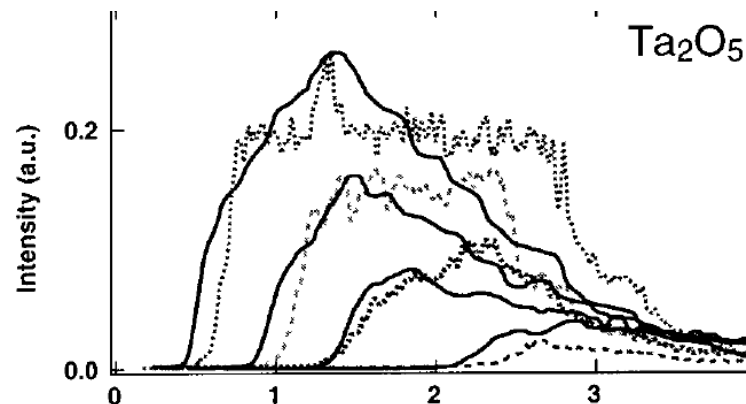
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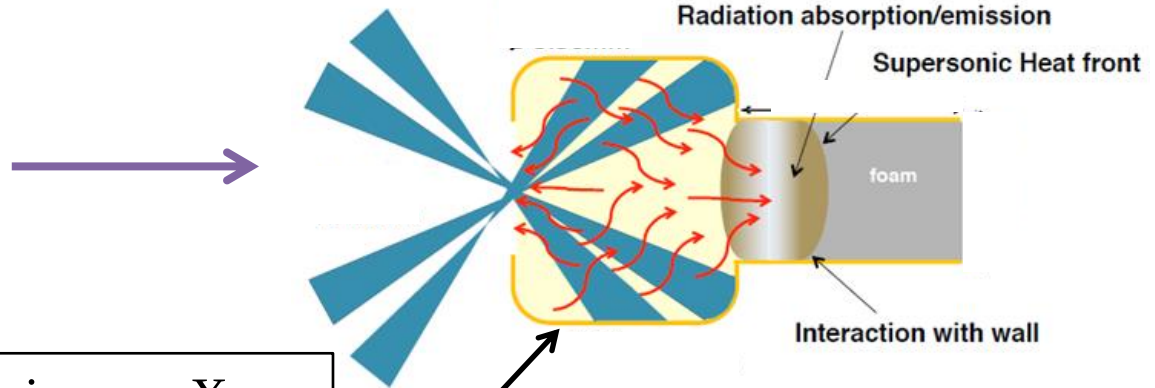
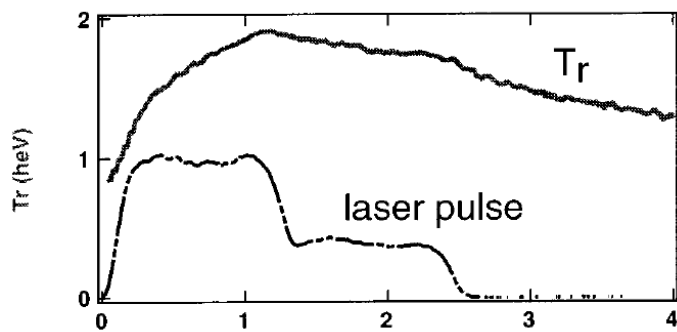
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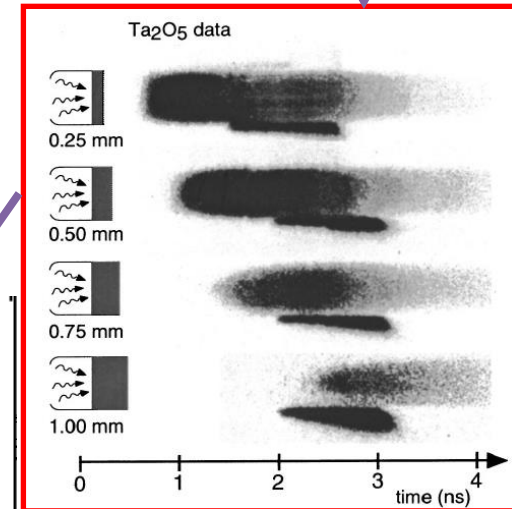
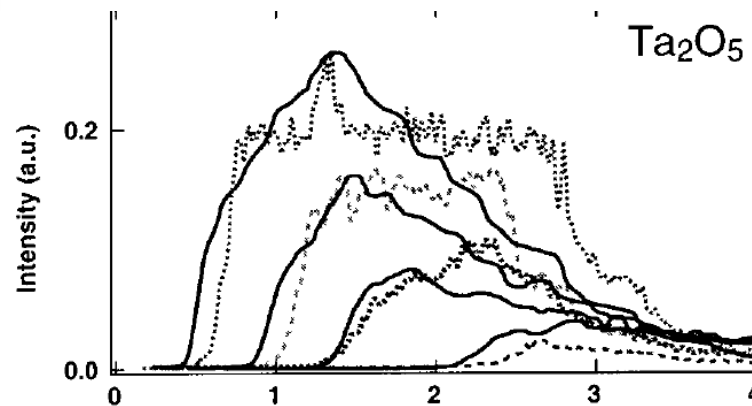
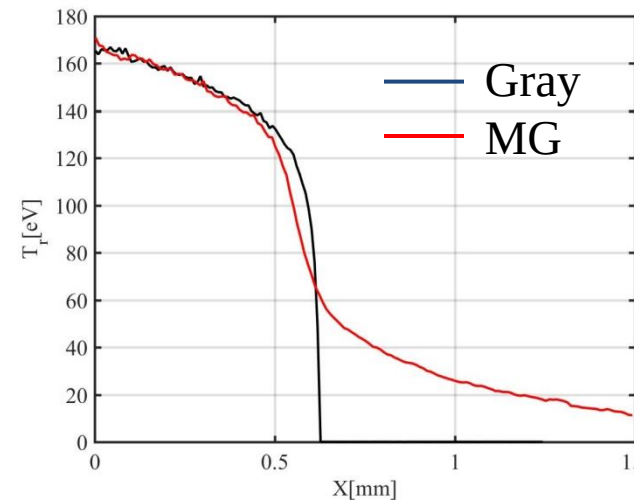
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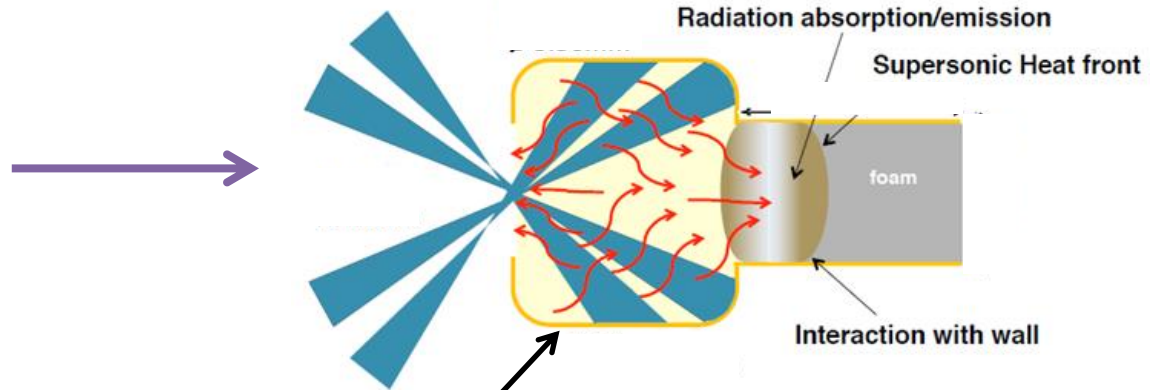
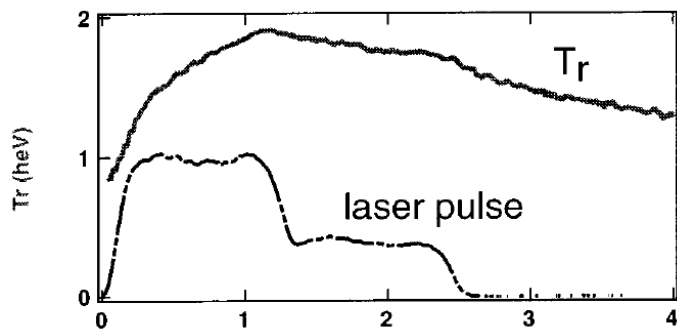
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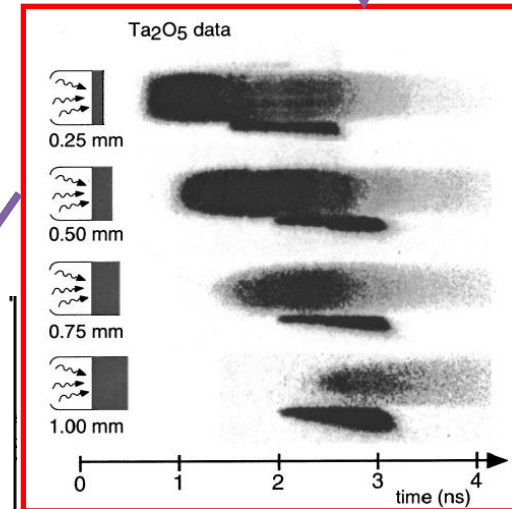
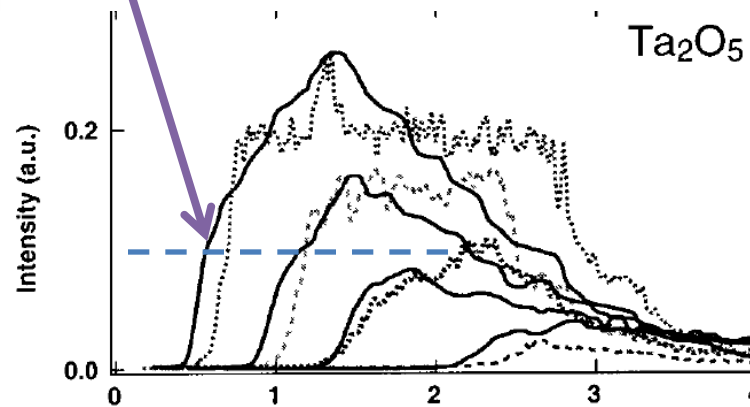
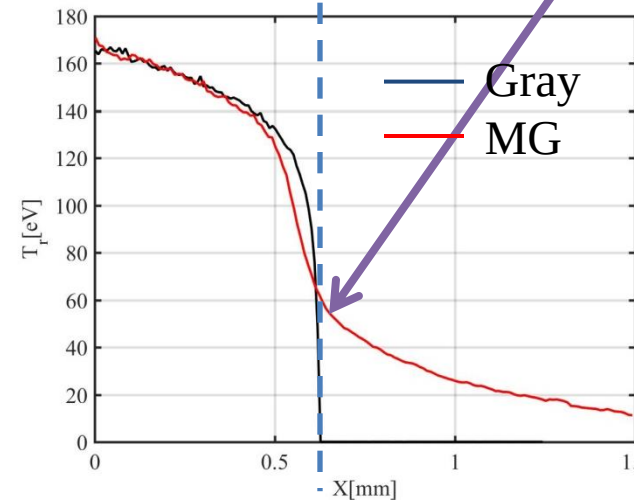
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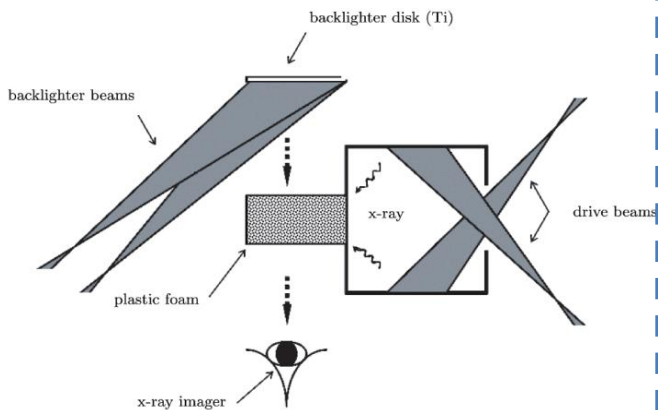


Breaking time

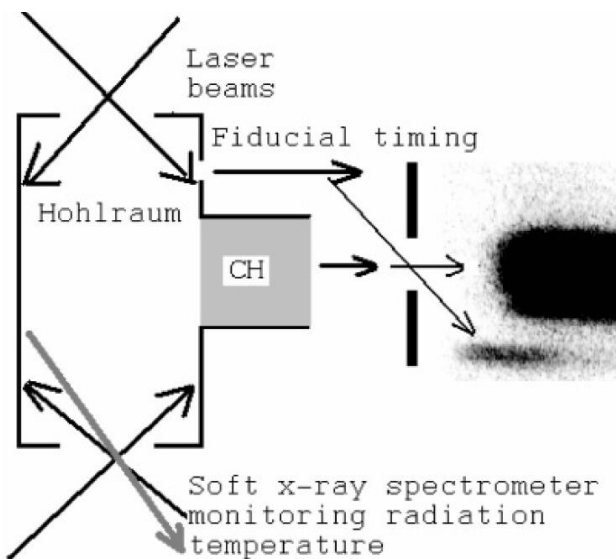


The Experiments Examined

SG-II

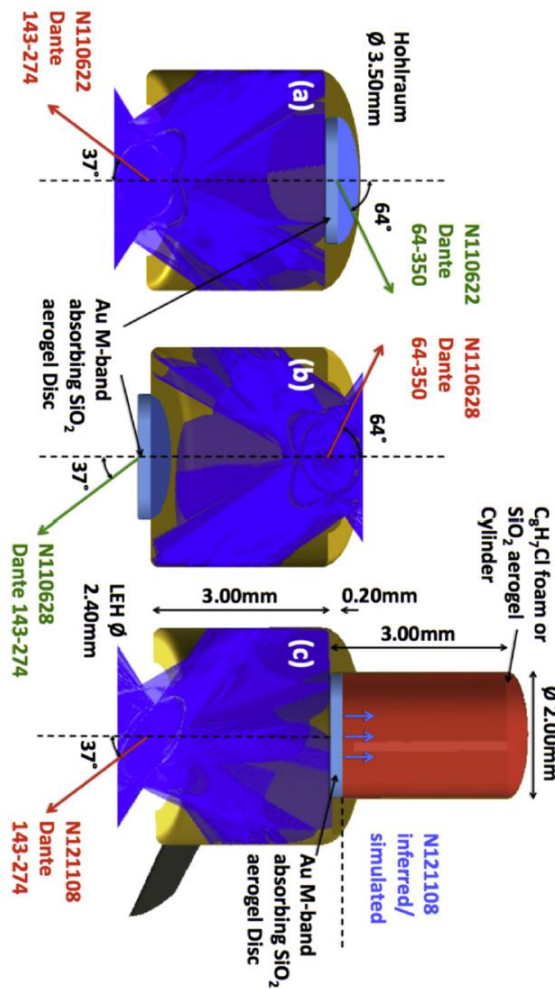


Zhang, *Chin. Phys. B*, 2010



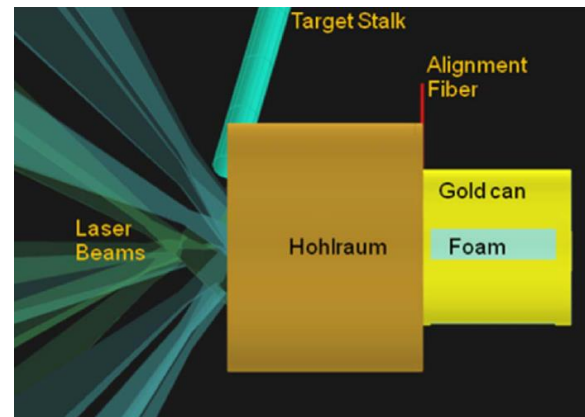
Xu, *Laser and Particle Beams*, 2006

NIF

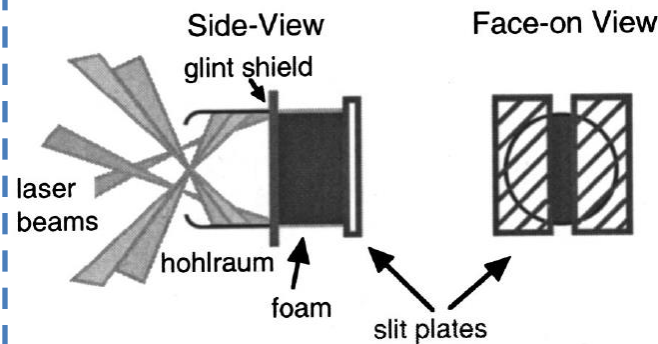


MOORE, *JQSRT*, 2015

OMEGA



Keiter, *Phys. Plas.*, 2008



Back, *PRL, Phys. Plas.*, 2000

Simple Analytic Models (based on Hammer & Rosen)

- The 1D diffusion approximation (hydrodynamics is negligible).
- Local Thermodynamic Equilibrium (LTE)
- Assuming:

opacity

$$\frac{1}{K} = gT^\alpha \rho^{-\lambda}$$

Internal energy

$$e = fT^\beta \rho^{-\mu}$$

The heat wave front in time (dependent on T_S)

$$x_F^2(t) = \frac{2 + \varepsilon}{1 - \varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t') dt'$$

$$H(t) = T_S^{4+\alpha}(t)$$

$$\varepsilon = \frac{\beta}{4 + \alpha}$$

$$C = \frac{16}{4 + \alpha} \frac{g \sigma_{SB}}{3 f \rho^{2-\mu+\lambda}}$$

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What exactly is T_s ?

$$\varepsilon = \frac{\beta}{4 + \alpha}$$

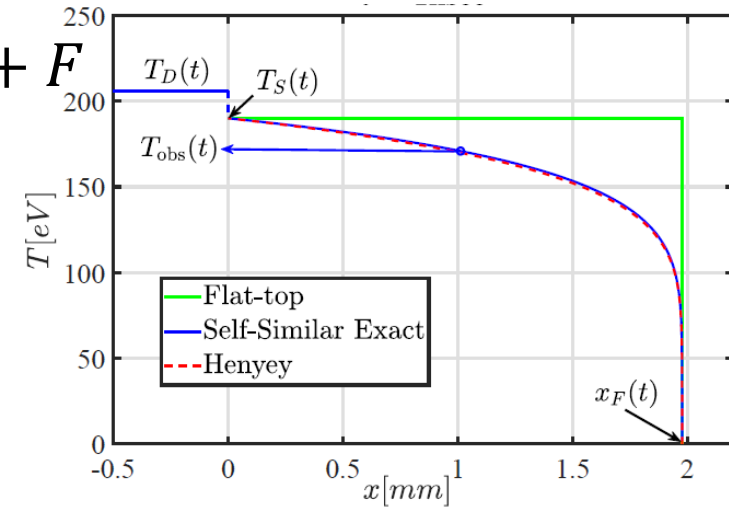
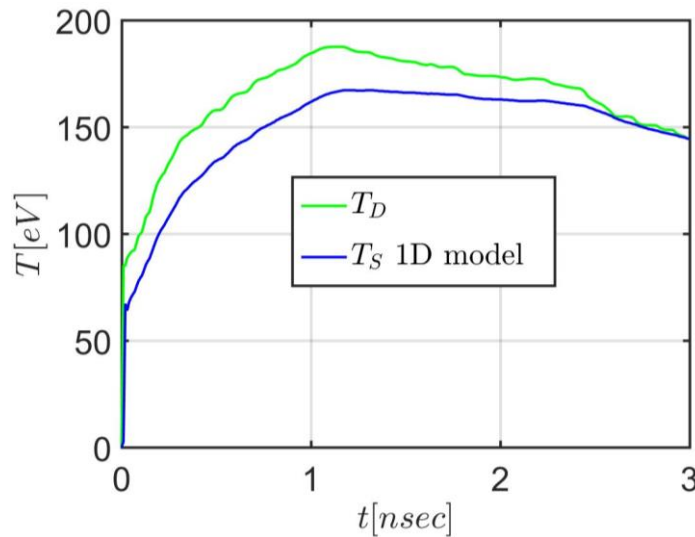
$$C = \frac{16}{4 + \alpha} \frac{g \sigma_{SB}}{3 f \rho^{2-\mu+\lambda}}$$

The 3 Radiation Temperatures

$$F_D = \sigma_{SB} T_D^4$$

$$\sigma_{SB} T_D^4 = \sigma_{SB} T_S^4 + F/2$$

$$\sigma_{SB} T_D^4 = \sigma_{SB} T_{OBS}^4 + F$$



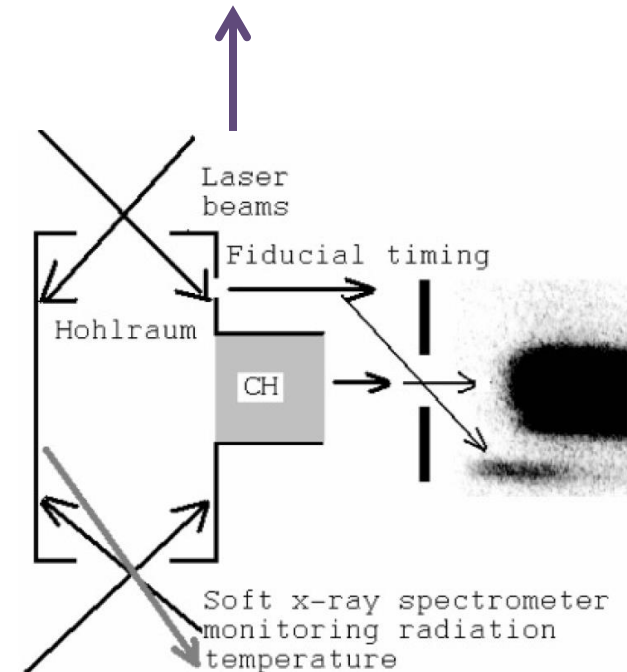
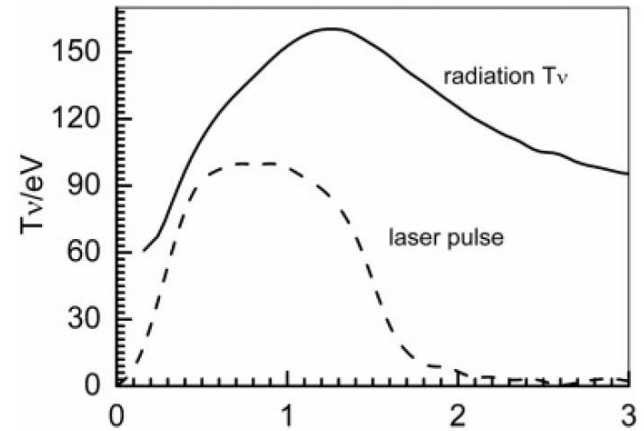
$$x_F^2(t) = \frac{2 + \varepsilon}{1 - \varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t') dt'$$

$$H(t) = T_S^{4+\alpha}(t)$$

What exactly is T_S ? T_S is the surface temperature not the driven temperature

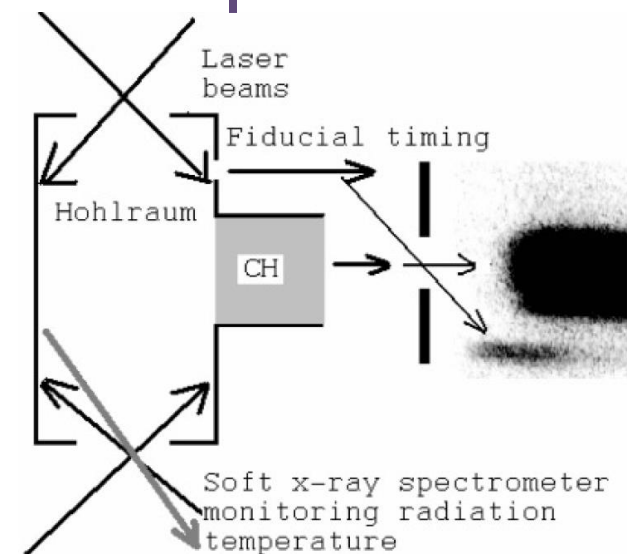
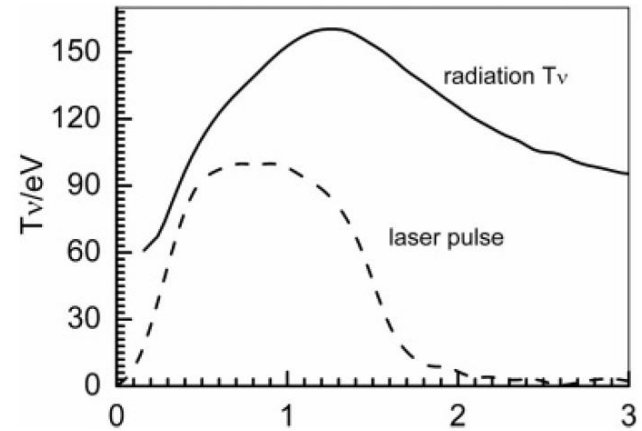
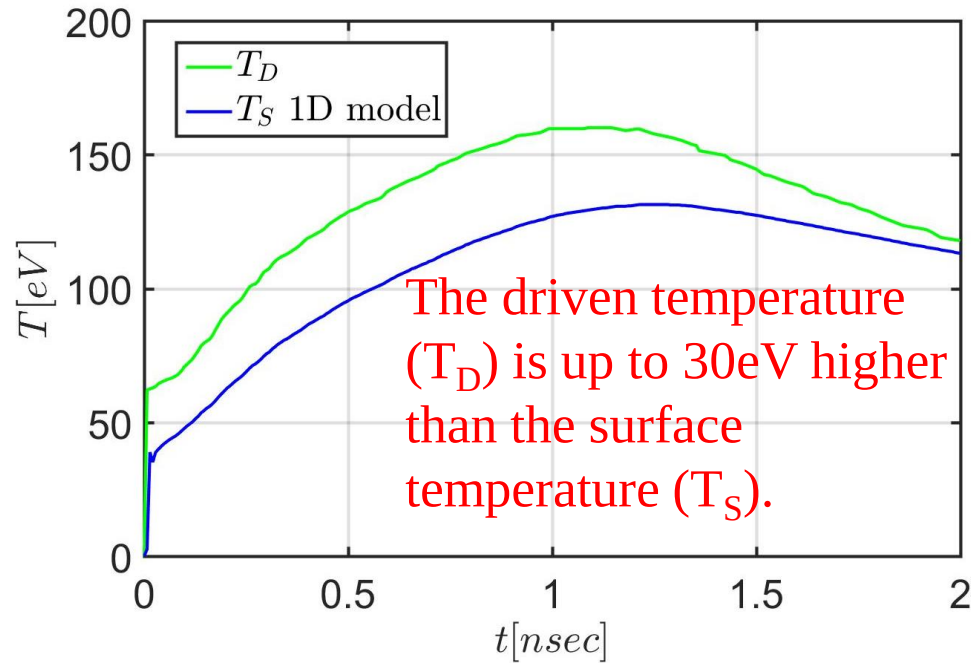
The 1D Analytic Model

Example SG-II Xu, *Laser and Particle Beams*, 2006 – C_6H_{12} Foam



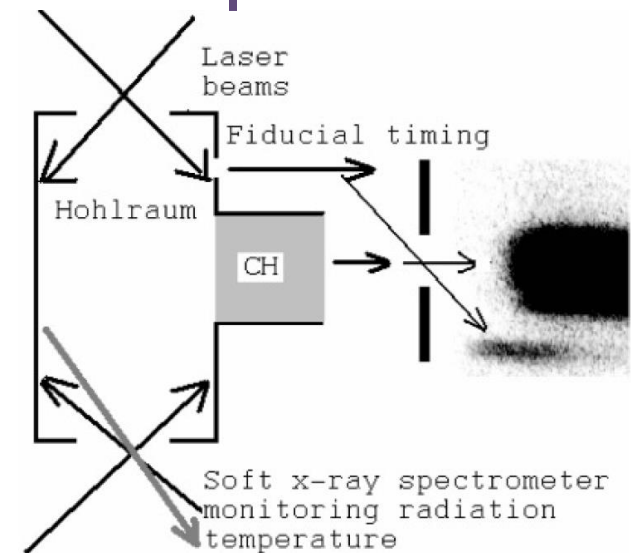
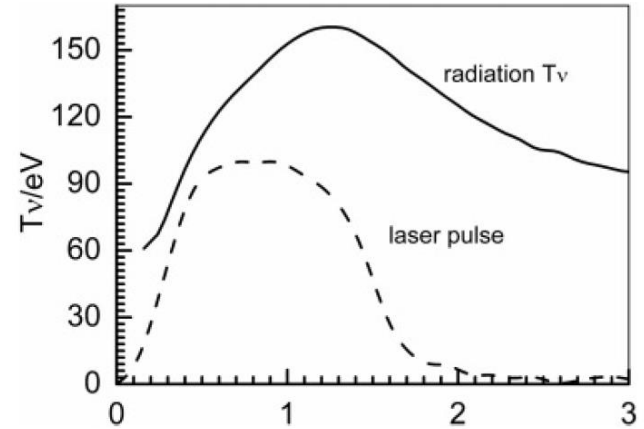
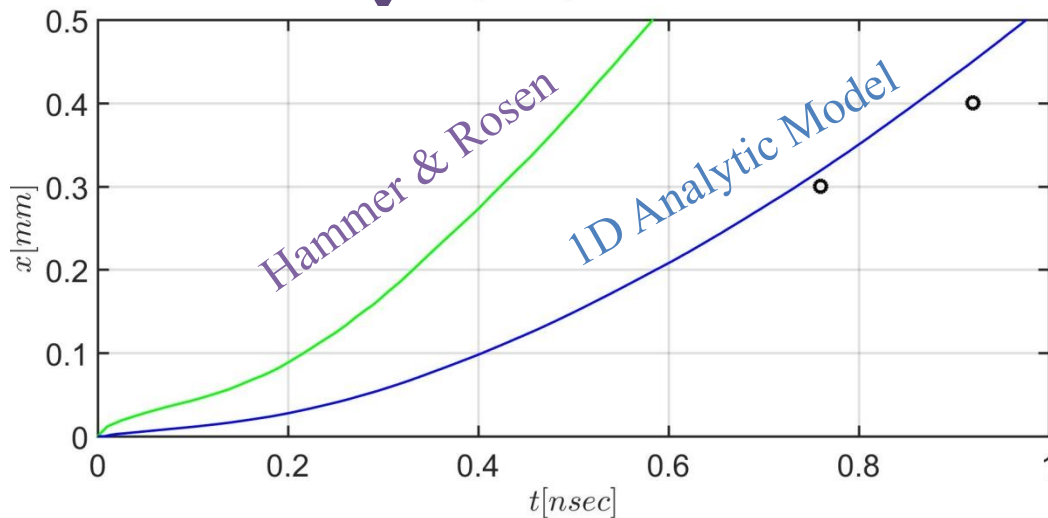
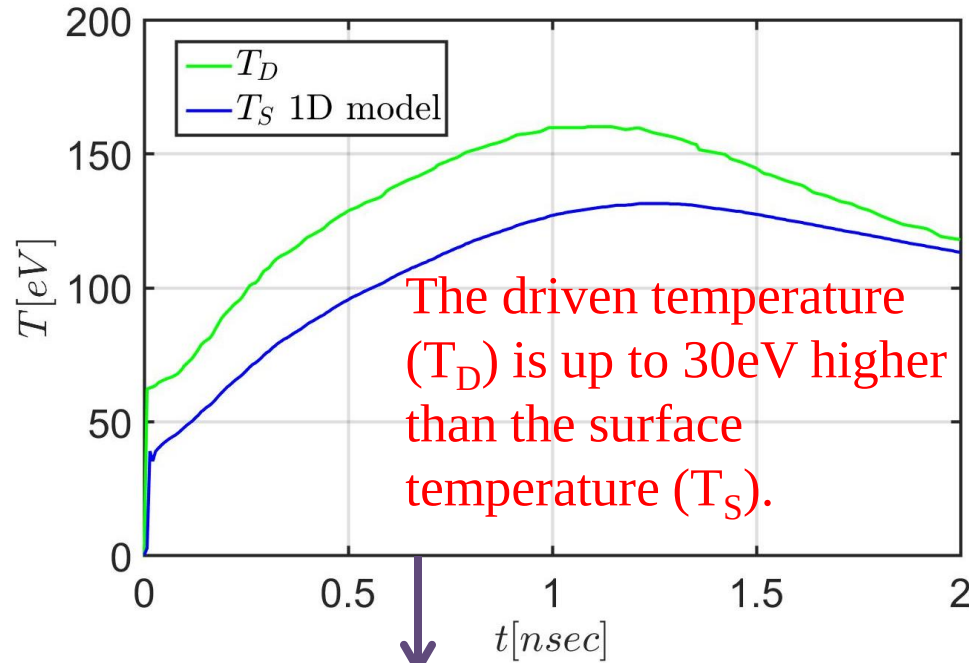
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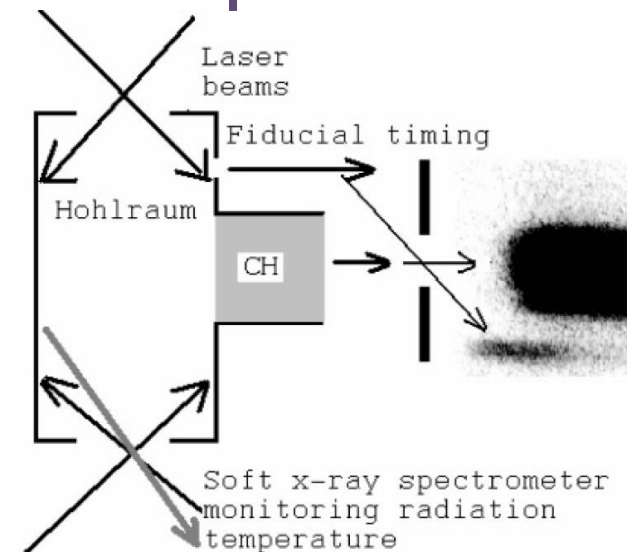
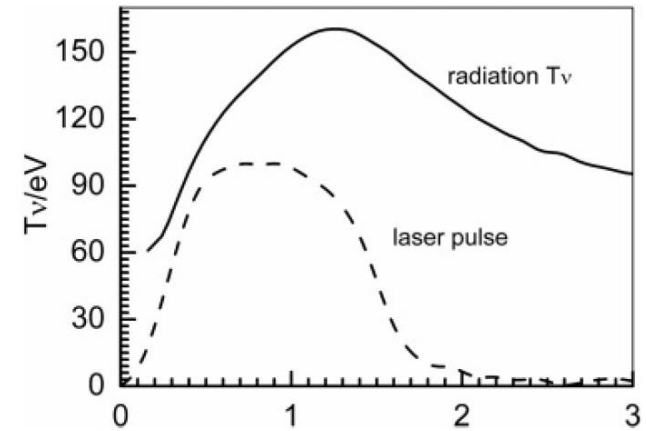
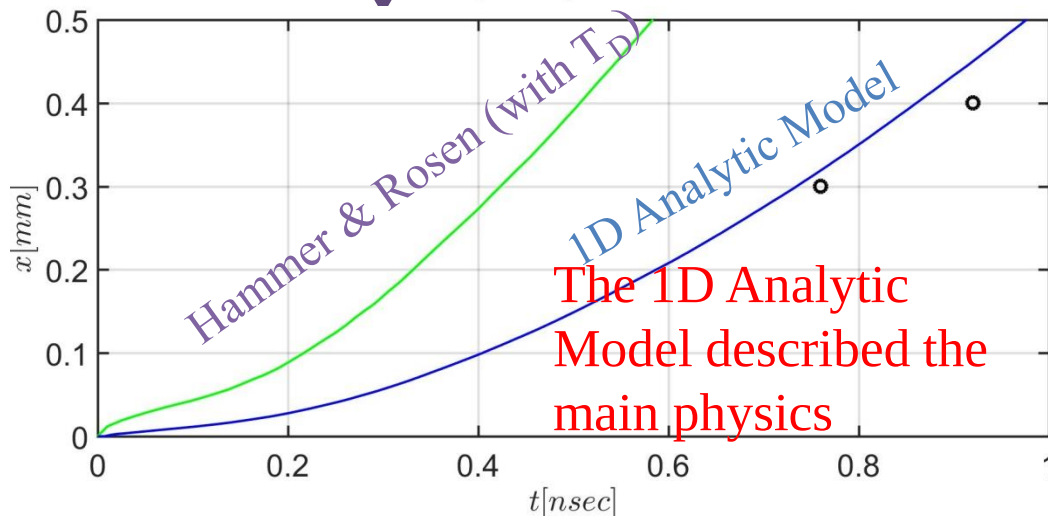
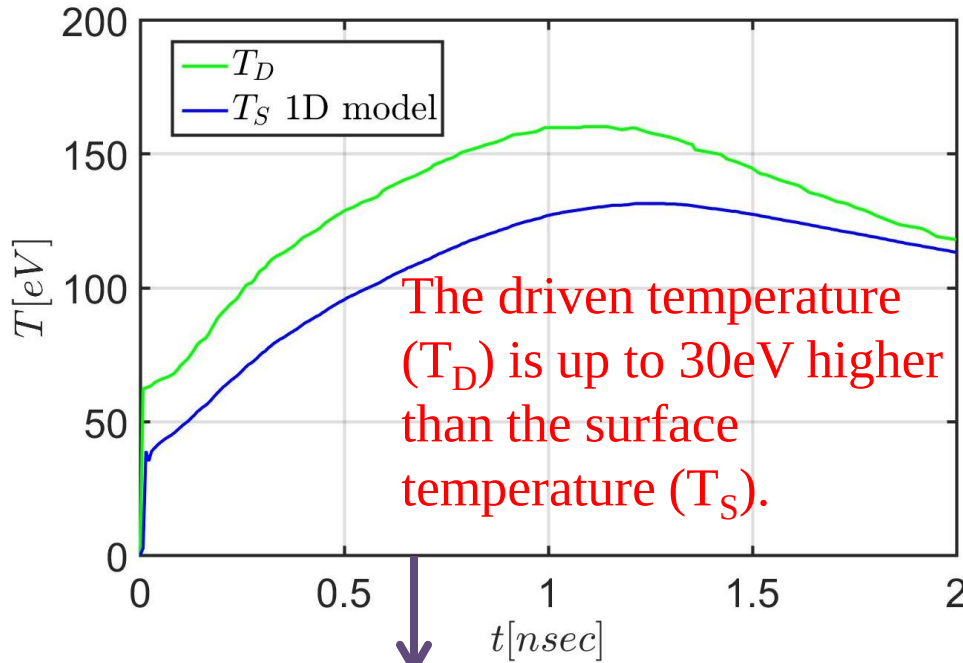
The 1D Analytic Model

Example SG-II Xu, *Laser and Particle Beams*, 2006 – C_6H_{12} Foam



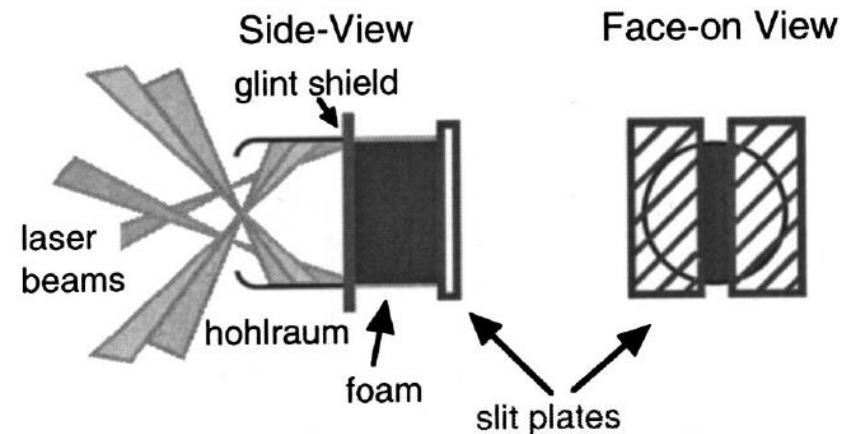
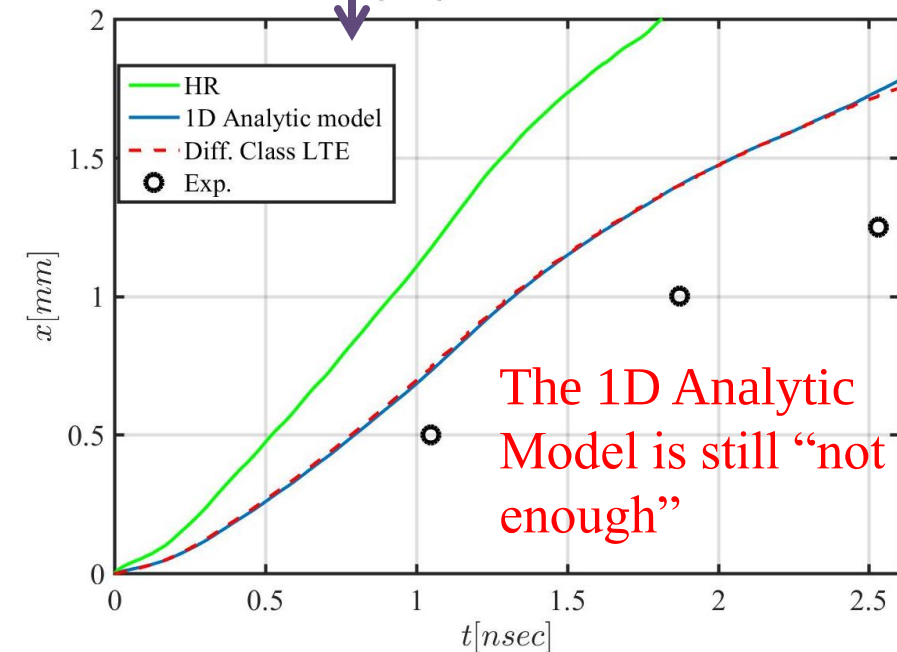
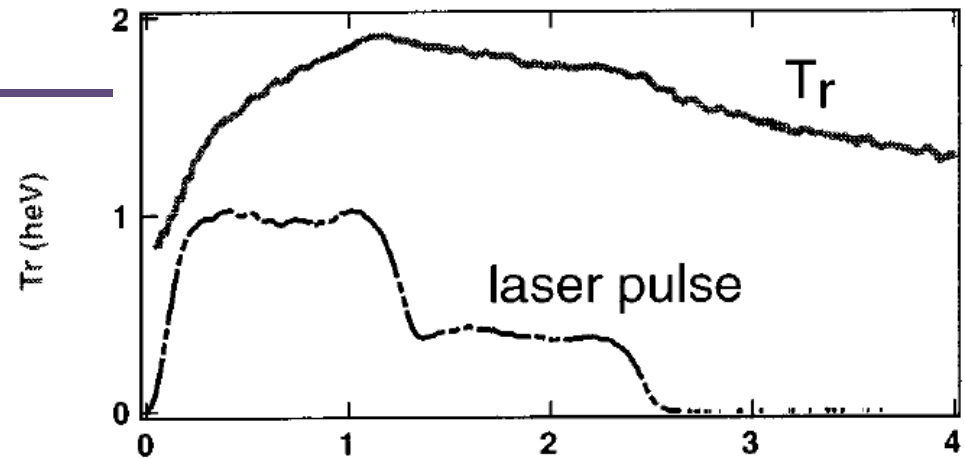
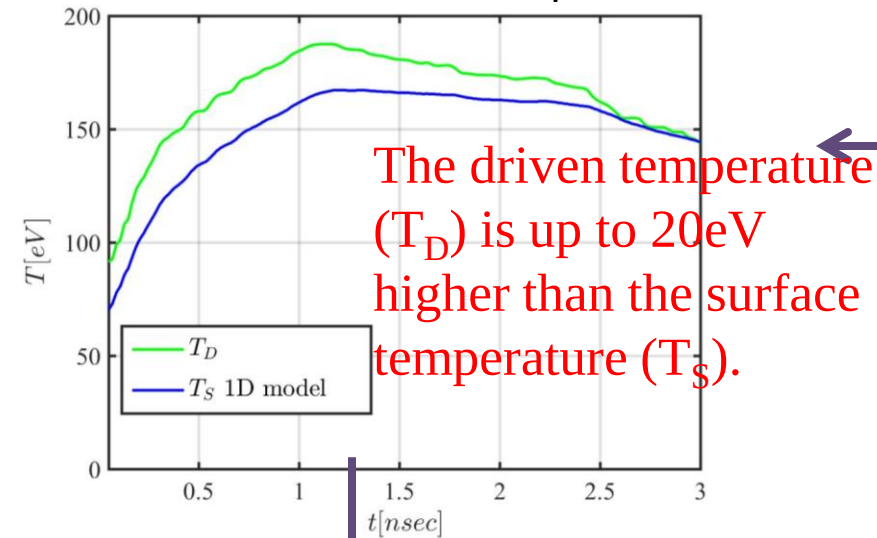
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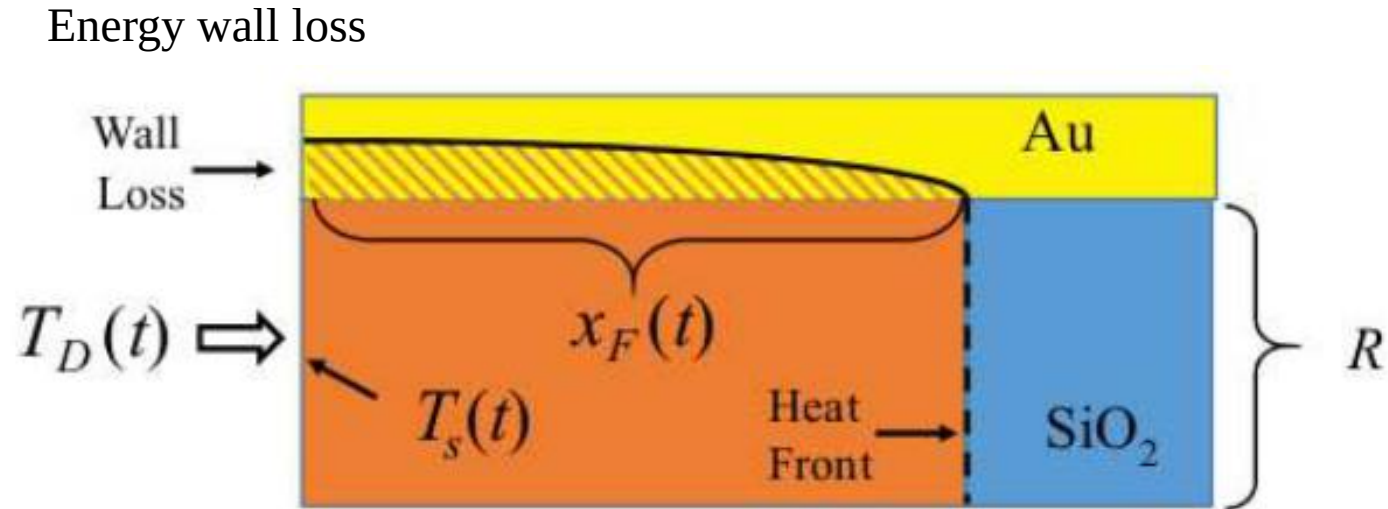
The 1D Analytic Model

Example OMEGA Back, *Phys. Plas.*, 2000 – SiO₂ Foam



Cohen & Heizler, *JCTT* (accepted)

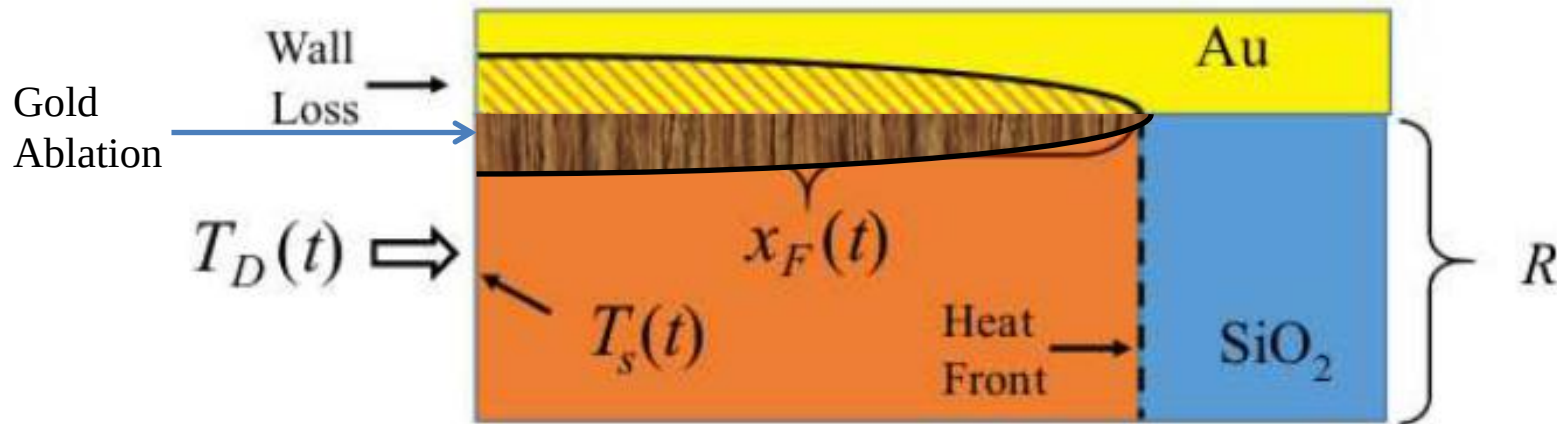
2D Effects



The heat wave loses energy to the gold walls, and therefore its velocity becomes slower

2D Effects

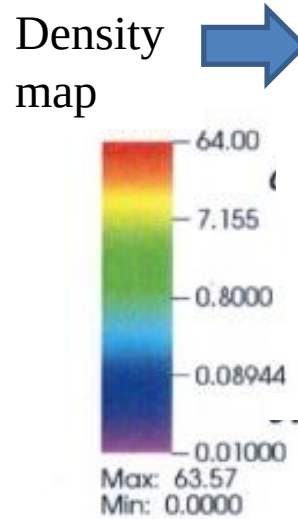
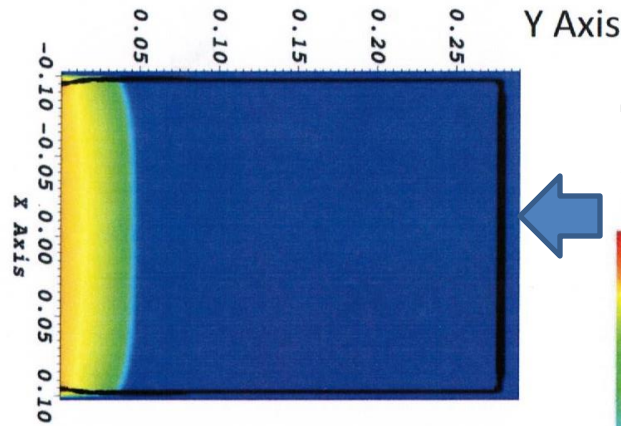
Energy wall loss



The heat wave loses energy to the gold walls, and therefore its velocity becomes slower

The gold is ablated into the foam and blocks a part of the energy and increases the foam's density

2D Effects - Demonstration via Exact Simulations

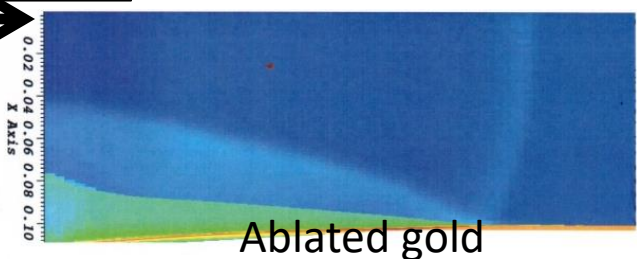
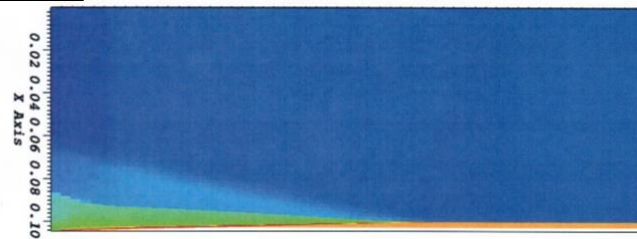
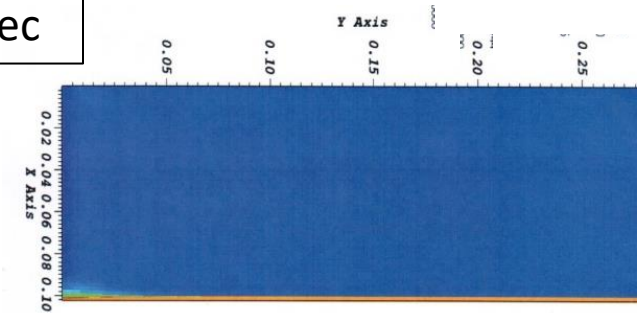


time=1nsec

time=2nsec

time=2.5nsec

Foam center

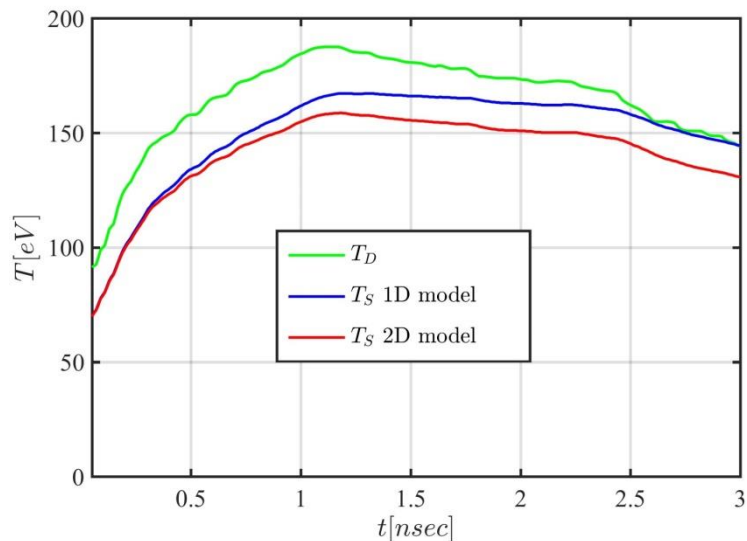


The 2D Analytic Model

Example OMEGA Back, *Phys. Plas.*, 2000 – SiO₂ Foam

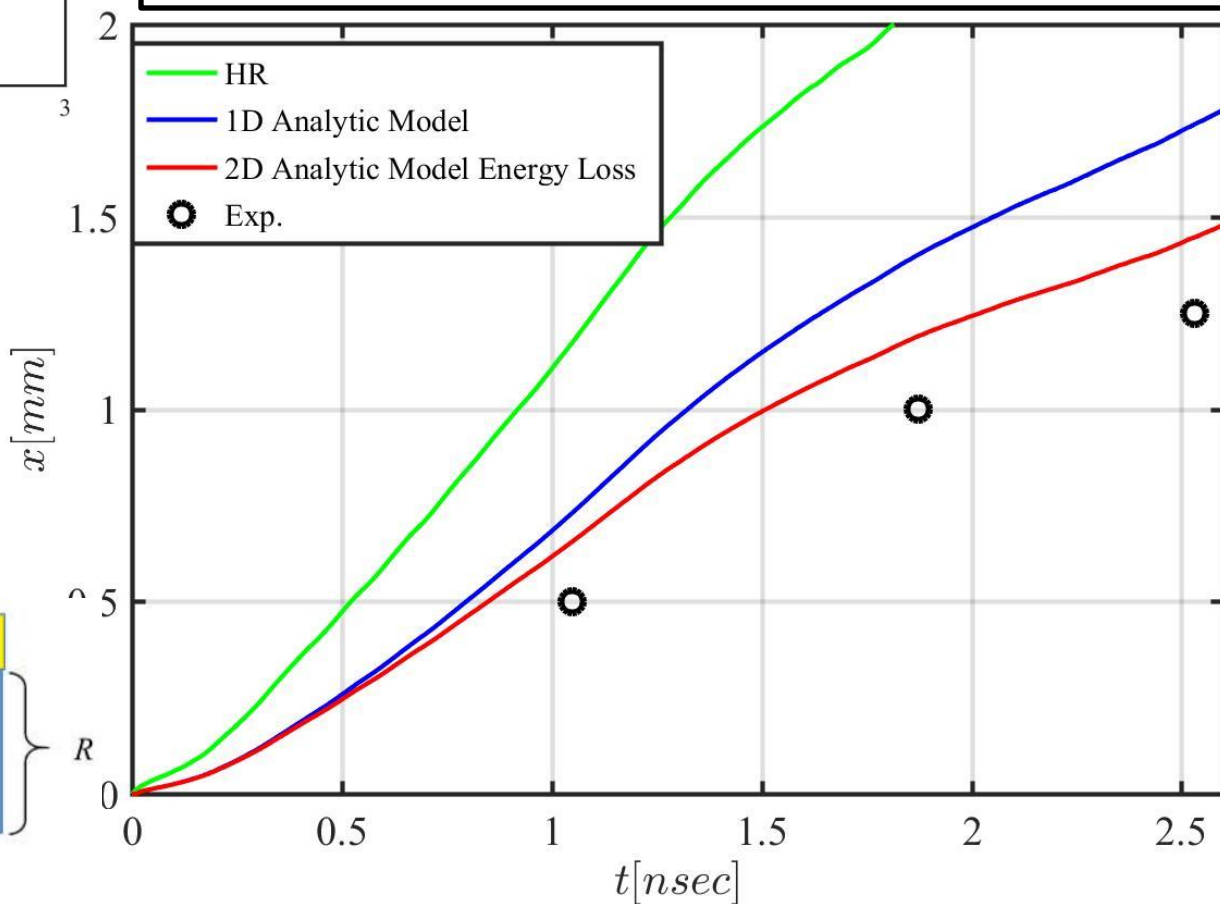
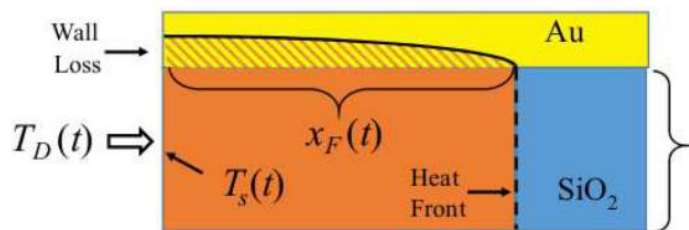
The energy which is transferred toward the gold can be taken from self similar subsonic analytic solutions.

The main physics is described by our 2D model



The energy losses to the walls decreases the surface temperature (T_S) in up to 10eV.

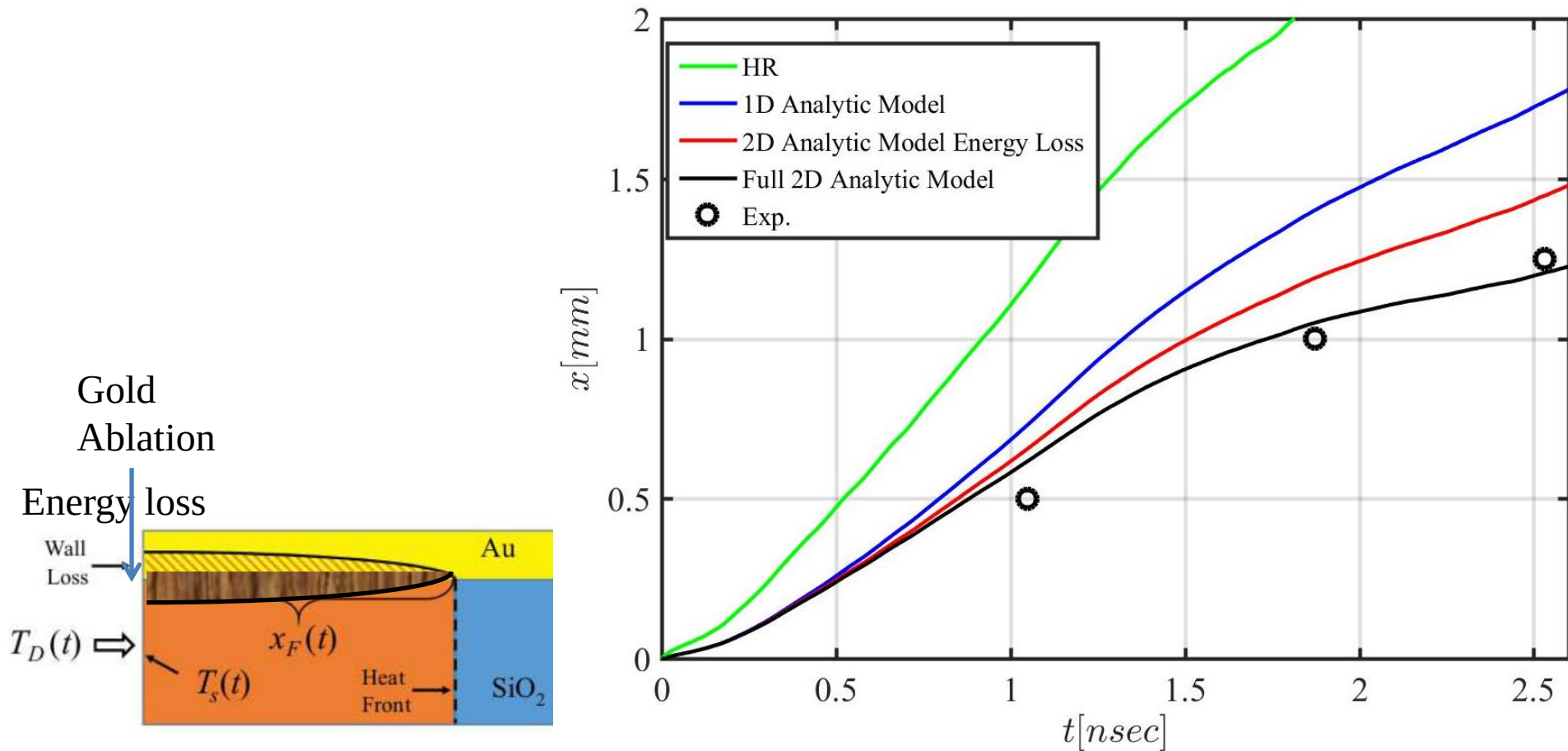
Energy loss



The Full 2D Analytic Model

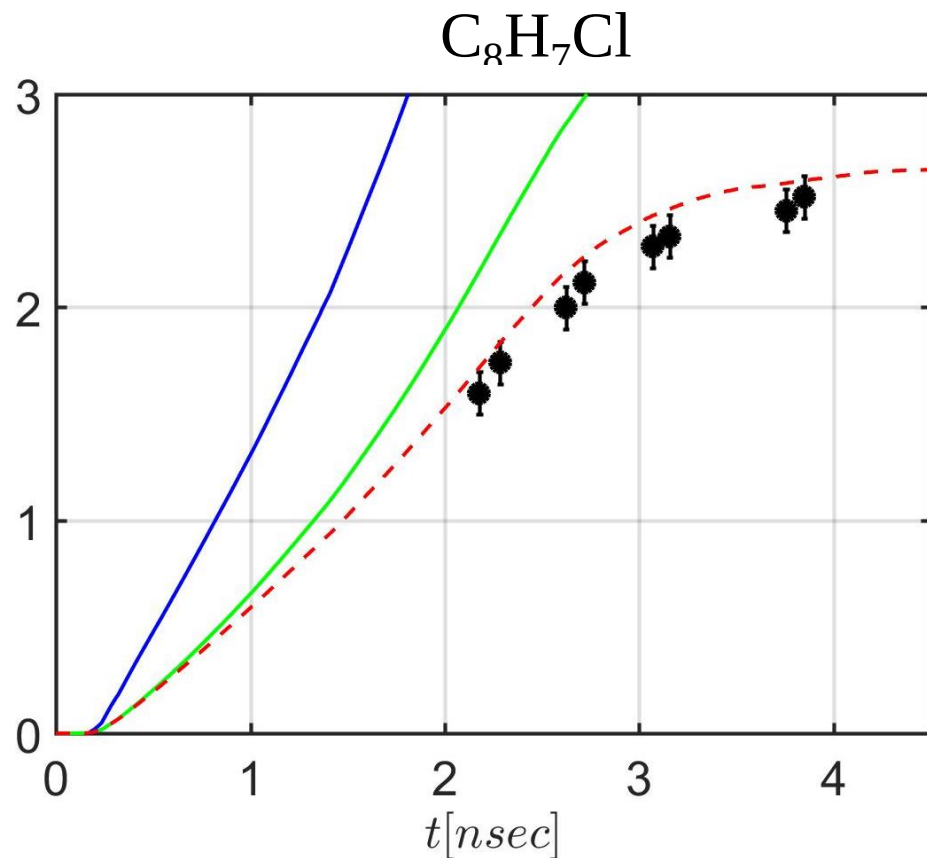
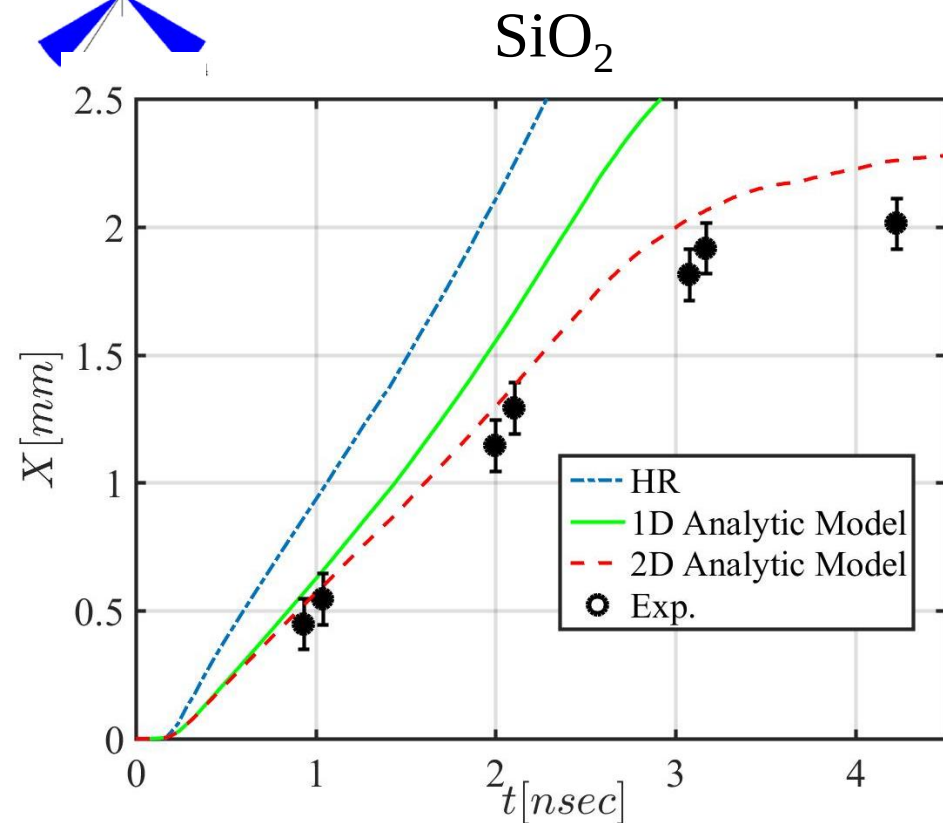
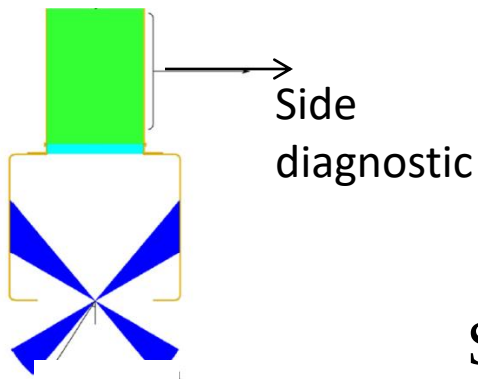
Example OMEGA Back, *Phys. Plas.*, 2000 – SiO₂ Foam

Taking the energy loss and the gold ablation into account
describing the experiment properly



MOORE (NIF) Experiment

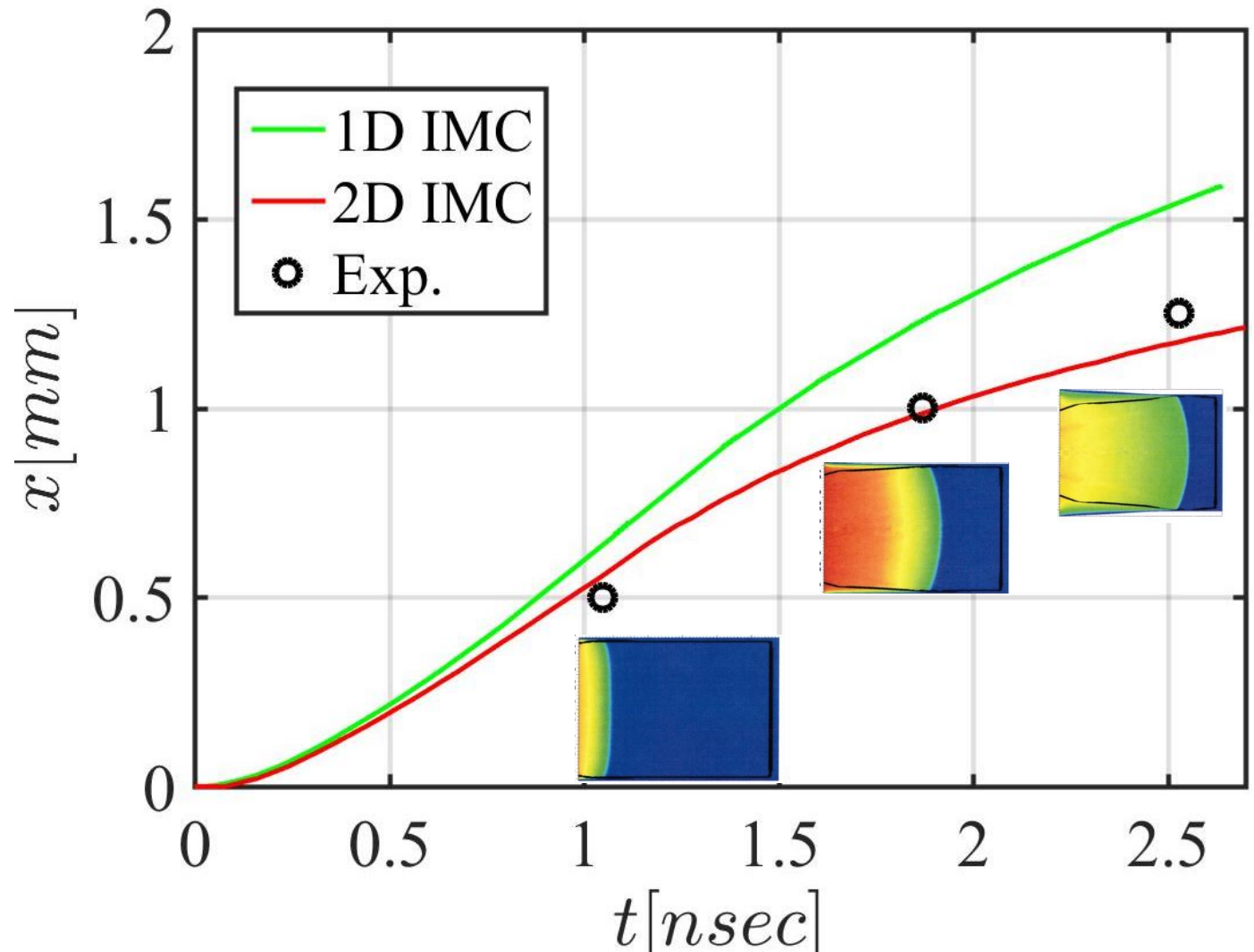
Moore, *JQSRT*, 2015



Advanced Simulations

OMEGA Back, *Phys. Plas.*, 2000 –SiO₂ Foam

The **2D**
simulation
are necessary

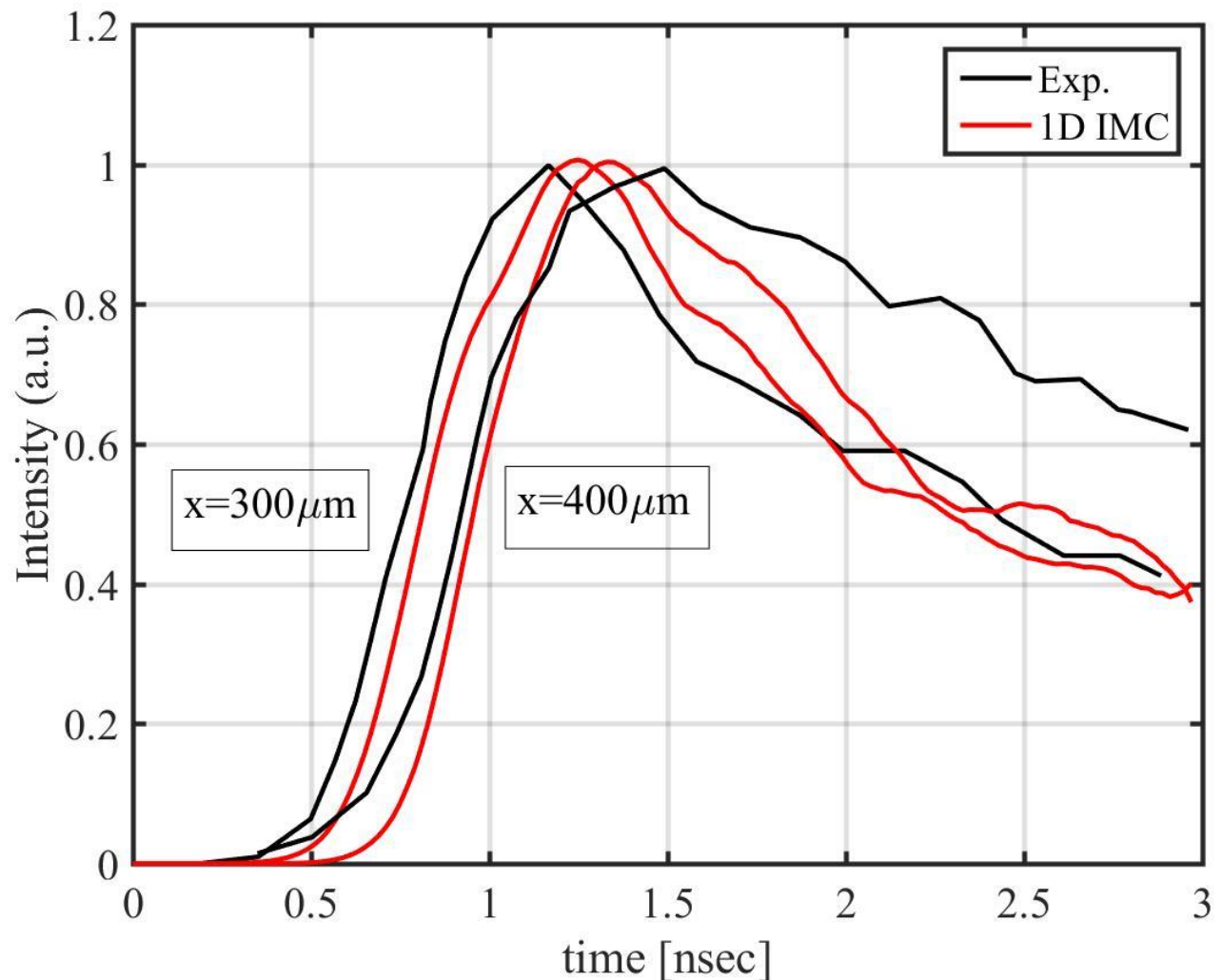


Advanced Simulations

SG-II Xu, *Laser and Particle Beams*, 2006 – C_6H_{12} Foam

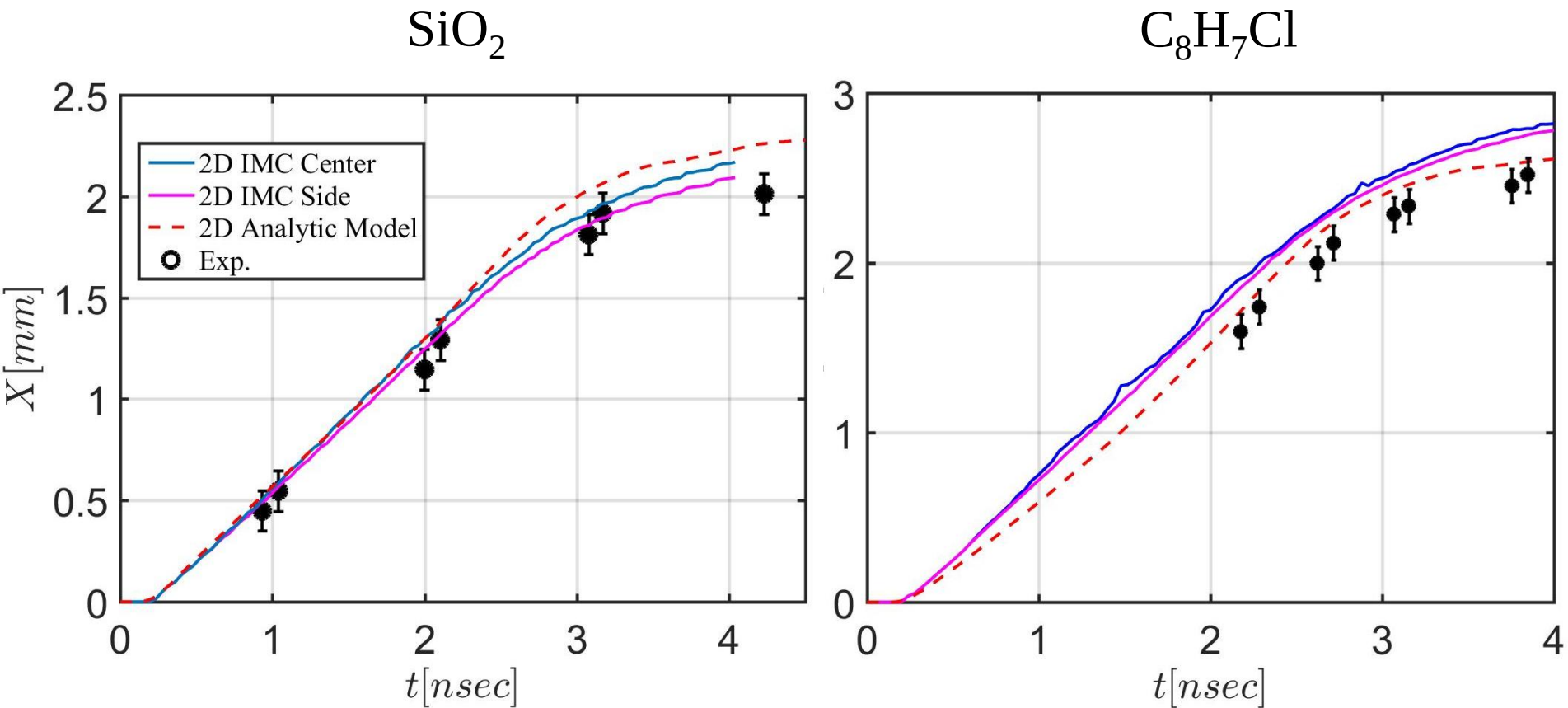
Comparing the flux in the edge of the sample, 1D simulations matches the experimental results.

C_6H_{12} in two different lengths



MOORE (NIF) Experiment Simulations & Analytic Model

Moore, *JQSRT*, 2015



Summary

- Supersonic Marshak heat wave experiments can be chiefly described by a simple analytic model;
- In optically thin systems, the experiments are well described by the 1D model & simulation
- When the opacity is higher, 2D effects should be taken into account

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Thanks ...

■ **Elad Malka**

■ **Guy Malamud** ■ **Roe Kirschenzweig**

■ **Roy Perry** ■ **PAZY Foundation, Grant No. 61139927**